Transfer-strategy results: thesis tables and figures

October 30, 2026

Table 1: Combination of TimesFM (ctx 1024), Chronos-2 (ctx 8192) and TabPFN-TS: aggregate relative MASE (↓), change vs TimesFM with 95% interval, and regret vs the ex-post best member per task. Evaluated windows 0–16 (0–6 for M4 and ILI); weights and selections use earlier windows only. Intervals: 95% task-stratified bootstrap (2,000 replicates) resampling series and windows.

Method	Score	% vs TimesFM [CI]	Mean regret	Max regret
NNLS, per task	0.633	−4.5 [−6.6, −2.5]	1.002	1.025
Inverse MASE^4, per series	0.633	−4.5 [−6.2, −2.7]	1.002	1.041
NNLS, per task × 3 horizon bins	0.634	−4.3 [−6.4, −2.3]	1.004	1.015
NNLS, per task × 6 horizon bins	0.635	−4.1 [−6.3, −2.1]	1.006	1.021
Inverse MASE^2, top 2, per series	0.637	−3.9 [−6.0, −2.0]	1.008	1.042
NNLS, per series	0.638	−3.7 [−5.8, −1.6]	1.011	1.052
Best member, per task	0.640	−3.5 [−5.6, −1.5]	1.013	1.063
Best member, per series	0.642	−3.1 [−5.2, −1.2]	1.016	1.052
Inverse MASE^2, per series	0.644	−2.9 [−5.3, −0.4]	1.020	1.136
NNLS, per task × 3 bins, 50% shrinkage	0.652	−1.6 [−4.6, +2.0]	1.037	1.346
Best member, per series, last window	0.653	−1.5 [−3.9, +0.9]	1.034	1.111
NNLS, global	0.657	−0.9 [−2.2, +1.2]	1.041	1.089
Inverse MASE, per series	0.660	−0.4 [−3.2, +3.0]	1.052	1.451
FFORMA-style meta-learner (LODO)	0.661	−0.3 [−2.5, +3.3]	1.046	1.094
TimesFM	0.663	–	1.050	1.101
Equal mean	0.682	+2.9 [−0.5, +7.5]	1.102	1.939
Median	0.684	+3.2 [−0.3, +7.8]	1.120	2.256
Chronos-2	0.702	+5.9 [+1.4, +11.1]	1.181	2.899
TabPFN-TS	0.773	+16.6 [+11.1, +24.1]	1.287	2.919

Table 2: Error redundancy of the zero-shot foundation models (squared scaled errors). Shared: share of member error explained by the members' mean error; 1st PC: share of the first principal component. Change in MSE vs the best member for ex-post optimal simplex weights (ideal), equal weights, and ideal weights with SeasonalNaive, AutoETS and AutoTheta added (+stats). TimesFM–Chronos-2 pair: uncentred error correlation ρ, accuracy ratio $k = \sigma_1/\sigma_2$ and the Bates–Granger optimum $(1-\rho^2)/(1+k^2-2\rho k)$, which is below one only if $\rho < k$. Evaluated windows 0–16 (0–6 for M4 and ILI); weights and selections use earlier windows only.

Task	Shared	1st PC	Best	Ideal	Equal	Ideal +stats	ρ	k	Pair opt.
M4 Hourly	0.85	0.84	TimesFM	−8.2	−7.2	−8.4	0.79	0.93	−5.1
ETTh1/48	0.94	0.94	TabPFN-TS	−3.3	−2.7	−6.2	0.90	0.97	−2.4
ETTh1/96	0.94	0.94	TabPFN-TS	−3.1	−1.4	−6.9	0.92	0.99	−3.3
ETTh2/48	0.93	0.93	Chronos-2	−1.2	+1.5	−1.4	0.94	0.94	0.0
ETTm1/48	0.89	0.89	Chronos-2	−11.6	−4.3	−14.7	0.91	0.90	0.0
ETTm2/48	0.90	0.90	Chronos-2	−1.9	+2.4	−6.2	0.91	0.97	−1.8
ECL	0.90	0.91	Chronos-2	−4.3	+1.1	−4.3	0.90	0.99	−4.3
Traffic	0.91	0.92	TimesFM	−0.4	+7.9	−0.9	0.93	0.95	−0.4
Weather	0.94	0.94	Chronos-2	−2.4	−1.0	−15.3	0.93	0.96	−1.0
Exchange	0.92	0.92	Chronos-2	0.0	+9.3	−0.2	0.97	0.96	−0.2
ILI	0.60	0.86	TimesFM	−12.2	+206.1	−12.6	0.01	0.38	−12.2
Median	0.91	0.92		−3.1	+1.1	−6.2	0.91	0.96	−1.8

Table 3: TimesFM context length: change in relative MASE vs ctx 1024 (%, ↓), flip invariance on. First column: relative MASE at ctx 1024. No flip: ctx 16256 without flip invariance. Selected: context and flip setting chosen per task from earlier windows (12 variants). Chronos-2: ctx 8192, for reference. Bold: best context per task. Evaluated windows 0–16 (0–6 for M4 and ILI); weights and selections use earlier windows only.

Task	1024 (rel.)	512	1024	2048	4096	8192	16256	No flip	Selected	Chronos-2
M4 Hourly	0.578	+0.9	0	0.0	0.0	0.0	0.0	+3.2	0.0	+9.7
ETTh1/48	1.018	−0.8	0	−0.7	**−1.5**	−0.3	−1.1	+1.1	+0.7	−6.9
ETTh1/96	0.912	+0.5	0	−0.9	−1.3	**−1.4**	−1.3	−0.1	+1.2	−2.1
ETTh2/48	0.885	−2.4	0	−4.8	−4.6	**−4.9**	−4.6	−6.0	−5.0	−8.7
ETTm1/48	0.749	**−10.7**	0	+1.5	−0.7	−0.4	−0.2	−2.1	−6.1	−9.2
ETTm2/48	0.680	+1.6	0	+2.2	**−2.9**	−1.7	−1.5	−2.1	−2.4	−5.4
ECL	0.704	+0.7	0	+0.4	−0.8	+0.5	**−3.9**	−1.3	−3.7	−6.2
Traffic	0.523	+11.8	0	−5.4	−8.0	−10.8	**−11.5**	−9.7	−11.5	−0.5
Weather	0.451	+3.3	0	−2.9	−7.4	**−11.5**	−10.2	−7.8	−4.8	−8.9
Exchange	0.924	+1.2	0	−1.0	−3.5	−3.9	**−3.9**	−2.9	−2.6	−2.7
ILI	0.292	+13.1	0	0.0	0.0	0.0	0.0	+1.5	+0.3	+189.9
Aggregate	0.663	+1.6	0	−1.1	−2.8	−3.2	−3.5	−2.5	−3.2	+5.9

Table 4: Effect of turning flip invariance off: change in TimesFM relative MASE (%) vs flip on at the same context. Evaluated windows 0–16 (0–6 for M4 and ILI); weights and selections use earlier windows only.

Task	512	1024	2048	4096	8192	16256
M4 Hourly	+2.8	+3.2	+3.2	+3.2	+3.2	+3.2
ETTh1/48	+1.2	−1.7	+1.6	+3.0	+2.8	+2.2
ETTh1/96	+1.7	+0.3	+1.9	+1.8	+1.3	+1.2
ETTh2/48	+0.8	−0.5	−2.1	−1.5	−0.4	−1.5
ETTm1/48	+1.9	−1.7	−0.1	+0.2	−0.7	−1.9
ETTm2/48	+1.7	+2.9	+0.8	−1.0	+0.4	−0.7
ECL	+2.6	+2.9	+4.6	+3.8	+4.1	+2.7
Traffic	+0.6	+1.9	+3.0	+2.2	+1.9	+2.0
Weather	+1.6	+1.9	+2.6	+0.8	+1.5	+2.7
Exchange	+0.3	+1.7	+0.6	−0.1	+1.0	+1.0
ILI	+7.2	+1.5	+1.5	+1.5	+1.5	+1.5

Table 5: Chronos-2 relative MASE (↓) by context length. Evaluated windows 0–16 (0–6 for M4 and ILI); weights and selections use earlier windows only.

Task	1024	2048	4096	8192
M4 Hourly	**0.634**	**0.634**	**0.634**	**0.634**
ETTh1/48	0.954	0.945	**0.944**	0.948
ETTh1/96	0.903	0.896	0.895	**0.893**
ETTh2/48	0.852	0.810	0.810	**0.808**
ETTm1/48	**0.662**	0.695	0.683	0.680
ETTm2/48	0.665	0.659	0.645	**0.643**
ECL	**0.656**	0.658	0.659	0.660
Traffic	**0.514**	0.515	0.517	0.521
Weather	0.458	0.418	0.414	**0.411**
Exchange	0.937	0.929	0.910	**0.900**
ILI	**0.845**	**0.845**	**0.845**	**0.845**
Aggregate	0.715	0.707	0.703	0.702

Table 6: LoRA fine-tuning per task: change in relative MASE (%, ↓) vs the zero-shot model at the same inference context. TimesFM: LoRA trained at ctx 1024 (r=4, all linear layers), evaluated at ctx 1024 (vs zero-shot 1024) and at ctx 16256 (vs zero-shot 16256) for seeds 0–2. Chronos-2: library-default LoRA trained at ctx 2048, evaluated at 2048 and 8192. Score columns: zero-shot relative MASE. *: validation selection kept the zero-shot model. Common evaluation set: windows 0–13 (0–3 for M4 and ILI); the three oldest windows, used for fine-tuning early stopping, are excluded from all fitting and scoring.

Task	TFM 16256	LoRA@1024	LoRA@16256 s0	s1	s2	Chr 8192	LoRA@2048	LoRA@8192
M4 Hourly	0.584	−2.5	−2.5	−1.8	−3.1	0.650	−0.4	−0.4
ETTh1/48	1.029	−5.9	−4.0	0.0*	0.0*	0.943	−0.8	+0.3
ETTh1/96	0.917	−1.7	−0.4	−2.3	−2.6	0.916	−2.0	−0.4
ETTh2/48	0.847	−3.4	−0.2	+0.2	−0.8	0.830	0.0*	0.0*
ETTm1/48	0.729	0.0*	0.0*	0.0*	−2.6	0.687	0.0*	0.0*
ETTm2/48	0.653	0.0*	0.0*	0.0*	0.0*	0.628	0.0*	0.0*
ECL	0.675	−8.2	−4.4	−4.2	−4.9	0.658	−0.2	−0.2
Traffic	0.481	−1.5	+0.5	+1.1	+0.1	0.539	−0.5	−0.2
Weather	0.386	−0.2	+1.1	+1.0	+0.6	0.393	+1.9	+0.8
Exchange	0.862	−2.7	+1.0	−1.4	+0.2	0.875	0.0*	0.0*
ILI	0.260	0.0*	0.0*	0.0*	0.0*	0.838	0.0*	0.0*
Aggregate	0.630	−2.4	−0.8	−0.7	−1.2	0.702	−0.2	0.0

Table 7: Joint comparison of transfer strategies: aggregate relative MASE (↓), change vs default TimesFM and vs TimesFM at ctx 16256 (% [95% CI]), GPU seconds per forecast origin summed over the 11 tasks, and one-time training GPU minutes (RTX A5000; TimesFM context capped at the longest history; adapters merged; TabPFN-TS from run logs). Combinations: task-level NNLS weights fitted on earlier windows; their cost is the sum of the members'. Common evaluation set: windows 0–13 (0–3 for M4 and ILI); the three oldest windows, used for fine-tuning early stopping, are excluded from all fitting and scoring. Intervals: 95% task-stratified bootstrap (2,000 replicates) resampling series and windows.

Strategy	Score	% vs default [CI]	% vs ctx 16256 [CI]	GPU s/origin	Train min
TimesFM, ctx 1024 (default)	0.655	–	+3.9 [+1.1, +6.4]	5.8	–
TimesFM, ctx 8192	0.633	−3.4 [−5.5, −1.0]	+0.4 [−0.4, +1.4]	37.4	–
TimesFM, ctx 16256	0.630	−3.8 [−6.0, −1.1]	–	72.7	–
TimesFM, context selected per task	0.630	−3.9 [−6.6, −1.5]	−0.1 [−2.0, +1.7]	197.1	–
TimesFM LoRA, inference ctx 1024	0.640	−2.4 [−4.0, −1.2]	+1.5 [−1.1, +3.8]	5.9	63
TimesFM LoRA, inference ctx 16256 (seed 0)	0.625	−4.6 [−7.1, −1.8]	−0.8 [−1.8, +0.2]	72.7	63
TimesFM LoRA, inference ctx 16256 (seed 1)	0.626	−4.5 [−7.0, −1.8]	−0.7 [−1.7, +0.2]	72.7	63
TimesFM LoRA, inference ctx 16256 (seed 2)	0.623	−5.0 [−7.5, −2.3]	−1.2 [−2.2, −0.3]	72.7	63
Chronos-2, ctx 8192	0.702	+7.2 [+2.5, +13.7]	+11.4 [+6.5, +18.0]	17.1	–
Chronos-2 LoRA, inference ctx 8192	0.702	+7.2 [+2.4, +13.7]	+11.4 [+6.5, +17.7]	18.2	175
TabPFN-TS	0.758	+15.7 [+9.4, +24.4]	+20.3 [+13.4, +29.0]	4079.5	–
TimesFM 1024 + Chronos-2 2048	0.633	−3.5 [−5.6, −1.5]	+0.4 [−2.4, +2.8]	10.5	–
TimesFM 1024 + Chronos-2 8192	0.629	−4.0 [−6.2, −1.8]	−0.2 [−2.9, +2.2]	23.0	–
TimesFM 1024 + Chronos-2 8192 + TabPFN-TS	0.629	−4.0 [−6.6, −1.5]	−0.3 [−3.0, +2.2]	4102.5	–
TimesFM 16256 + Chronos-2 8192	0.619	−5.5 [−8.1, −2.9]	−1.8 [−3.8, 0.0]	89.8	–
TimesFM LoRA@16256 + Chronos-2 8192	0.617	−5.8 [−8.6, −3.1]	−2.1 [−4.1, −0.1]	89.8	63
TimesFM LoRA@16256 + Chronos-2 LoRA@8192	0.618	−5.7 [−8.6, −2.8]	−2.0 [−4.1, +0.1]	90.9	239

Table 8: Combination gain (combination vs its TimesFM member, %) and differences between gains (ratio of ratios, %; positive: the gain is smaller than in the reference). Intervals: series and windows resampled, or series and circular blocks of three consecutive windows. Common evaluation set: windows 0–13 (0–3 for M4 and ILI); the three oldest windows, used for fine-tuning early stopping, are excluded from all fitting and scoring.

Quantity	% [CI], series × windows	% [CI], series × window blocks
Combination gain, TimesFM default + Chronos-2	−4.0 [−6.2, −1.8]	−4.0 [−6.2, −2.0]
Combination gain, TimesFM 16256 + Chronos-2	−1.8 [−3.8, 0.0]	−1.8 [−3.6, −0.3]
Combination gain, TimesFM LoRA@16256 + Chronos-2	−1.3 [−3.2, +0.5]	−1.3 [−3.0, +0.1]
Combination gain, TimesFM LoRA@16256 + Chronos-2 LoRA	−1.2 [−3.1, +0.7]	−1.2 [−3.0, +0.3]
Difference: 16256 vs default	+2.3 [+0.3, +4.0]	+2.3 [+0.3, +4.2]
Difference: LoRA@16256 vs 16256	+0.5 [0.0, +1.1]	+0.5 [+0.1, +1.1]
Difference: LoRA@16256 vs default	+2.8 [+0.8, +4.7]	+2.8 [+0.7, +4.9]
Difference: Chronos-2 LoRA added	+0.1 [−0.4, +0.6]	+0.1 [−0.3, +0.6]

Table 9: Adding TabPFN-TS as a third member: aggregate relative MASE (↓) of the TimesFM + Chronos-2 pair and of the pair plus TabPFN-TS (task-level NNLS weights), and the change from adding TabPFN-TS (% [95% CI]). Common evaluation set: windows 0–13 (0–3 for M4 and ILI); the three oldest windows, used for fine-tuning early stopping, are excluded from all fitting and scoring.

Combination	Pair	+ TabPFN-TS	Change, series × windows	Change, series × blocks
TimesFM ctx 1024 + Chronos-2	0.629	0.629	−0.1 [−1.1, +1.0]	−0.1 [−0.9, +0.9]
TimesFM ctx 16256 + Chronos-2	0.619	0.620	+0.1 [−0.7, +1.0]	+0.1 [−0.6, +1.0]
TimesFM LoRA@16256 + Chronos-2	0.617	0.618	+0.1 [−0.7, +1.0]	+0.1 [−0.6, +0.9]

Table 10: Relative MASE (↓) per task and strategy. Bold: best per task. TFM: TimesFM, ctx 1024 (default); TFM-L: TimesFM, ctx 16256; TFM-sel: TimesFM, context selected per task; TFM-LoRA: TimesFM LoRA, inference ctx 1024; TFM-L-LoRA: TimesFM LoRA, inference ctx 16256 (seed 0); Chr: Chronos-2, ctx 8192; Chr-LoRA: Chronos-2 LoRA, inference ctx 8192; TFM+Chr2k: TimesFM 1024 + Chronos-2 2048; TFM+Chr: TimesFM 1024 + Chronos-2 8192; TFM-L+Chr: TimesFM 16256 + Chronos-2 8192; TFM-L-LoRA+Chr: TimesFM LoRA@16256 + Chronos-2 8192. Common evaluation set: windows 0–13 (0–3 for M4 and ILI); the three oldest windows, used for fine-tuning early stopping, are excluded from all fitting and scoring.

Task	*TFM*	*TFM-L*	*TFM-sel*	*TFM-LoRA*	*TFM-L-LoRA*	*Chr*	*Chr-LoRA*	*TFM+Chr2k*	*TFM+Chr*	*TFM-L+Chr*	*TFM-L-LoRA+Chr*
M4 Hourly	0.584	0.584	0.584	0.569	0.569	0.650	0.648	0.571	0.571	0.571	**0.565**
ETTh1/48	1.044	1.029	1.050	0.983	0.988	**0.943**	0.946	0.983	0.982	0.989	0.973
ETTh1/96	0.924	0.917	0.926	**0.908**	0.913	0.916	0.913	0.918	0.914	0.918	0.914
ETTh2/48	0.886	0.847	0.853	0.856	0.845	0.830	0.830	**0.828**	0.831	0.830	0.831
ETTm1/48	0.743	0.729	0.689	0.743	0.729	**0.687**	**0.687**	0.722	0.708	0.720	0.720
ETTm2/48	0.658	0.653	0.644	0.658	0.653	**0.628**	**0.628**	0.638	0.629	0.629	0.629
ECL	0.706	0.675	0.684	0.648	**0.645**	0.658	0.656	0.671	0.666	0.658	0.646
Traffic	0.548	**0.481**	**0.481**	0.540	0.483	0.539	0.538	0.540	0.541	**0.481**	0.483
Weather	0.433	0.386	0.391	0.433	0.391	0.393	0.396	0.407	0.404	**0.385**	0.386
Exchange	0.899	**0.862**	0.862	0.875	0.871	0.875	0.875	0.895	0.890	**0.862**	0.871
ILI	0.260	0.260	0.260	0.260	0.260	0.838	0.838	0.250	0.250	**0.250**	0.250
Aggregate	0.655	0.630	0.630	0.640	0.625	0.702	0.702	0.633	0.629	0.619	0.617

Table 11: Per-task Diebold–Mariano tests on window-level MASE (Newey–West variance, HLN correction, Holm across the 11 tasks): tasks with lower mean loss / significantly lower / significantly higher at 5%. TimesFM ctx 16256 is a reference and is not tested against the default here. Common evaluation set: windows 0–13 (0–3 for M4 and ILI); the three oldest windows, used for fine-tuning early stopping, are excluded from all fitting and scoring.

Strategy	vs TimesFM default	vs TimesFM 16256
TimesFM, context selected per task	9 / 0 / 0	4 / 0 / 0
TimesFM LoRA, inference ctx 1024	10 / 1 / 0	5 / 0 / 0
TimesFM LoRA, inference ctx 16256 (seed 0)	11 / 0 / 0	6 / 1 / 0
Chronos-2, ctx 8192	9 / 0 / 1	6 / 0 / 1
Chronos-2 LoRA, inference ctx 8192	9 / 0 / 1	6 / 0 / 1
TimesFM 1024 + Chronos-2 2048	11 / 1 / 0	7 / 0 / 0
TimesFM 1024 + Chronos-2 8192	11 / 1 / 0	8 / 0 / 0
TimesFM 16256 + Chronos-2 8192	11 / 0 / 0	8 / 0 / 0
TimesFM LoRA@16256 + Chronos-2 8192	11 / 0 / 0	8 / 2 / 0

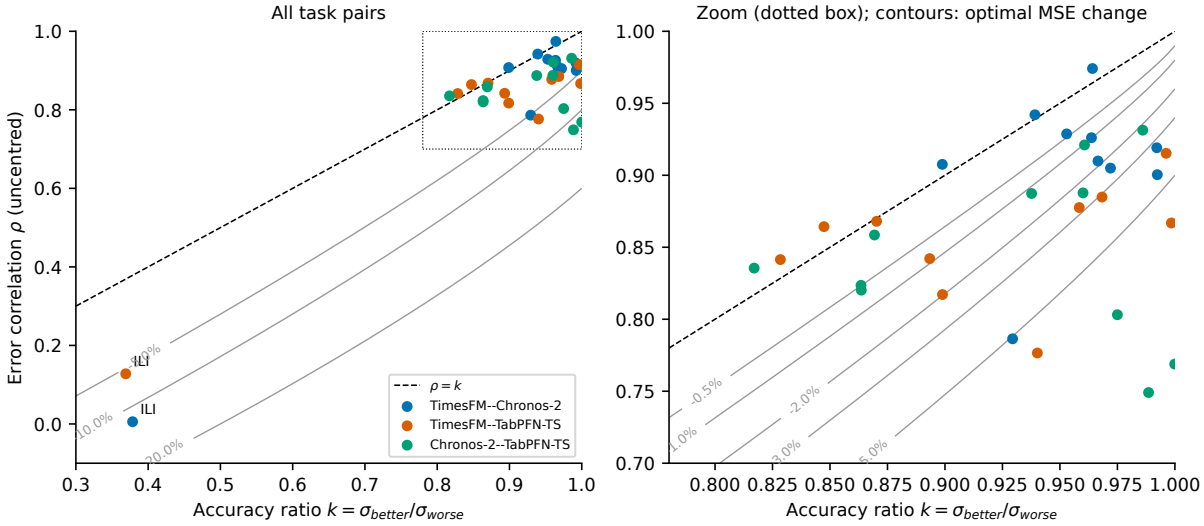


Figure 1: Task pairs on the correlation–accuracy plane with Bates–Granger iso-gain curves.

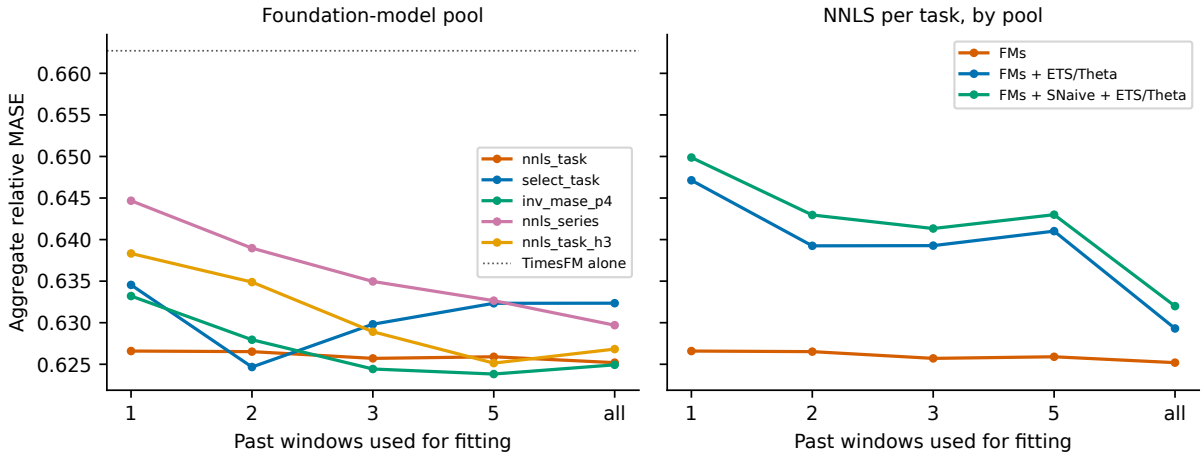


Figure 2: Combiner learning curves over the number of past windows used for fitting.

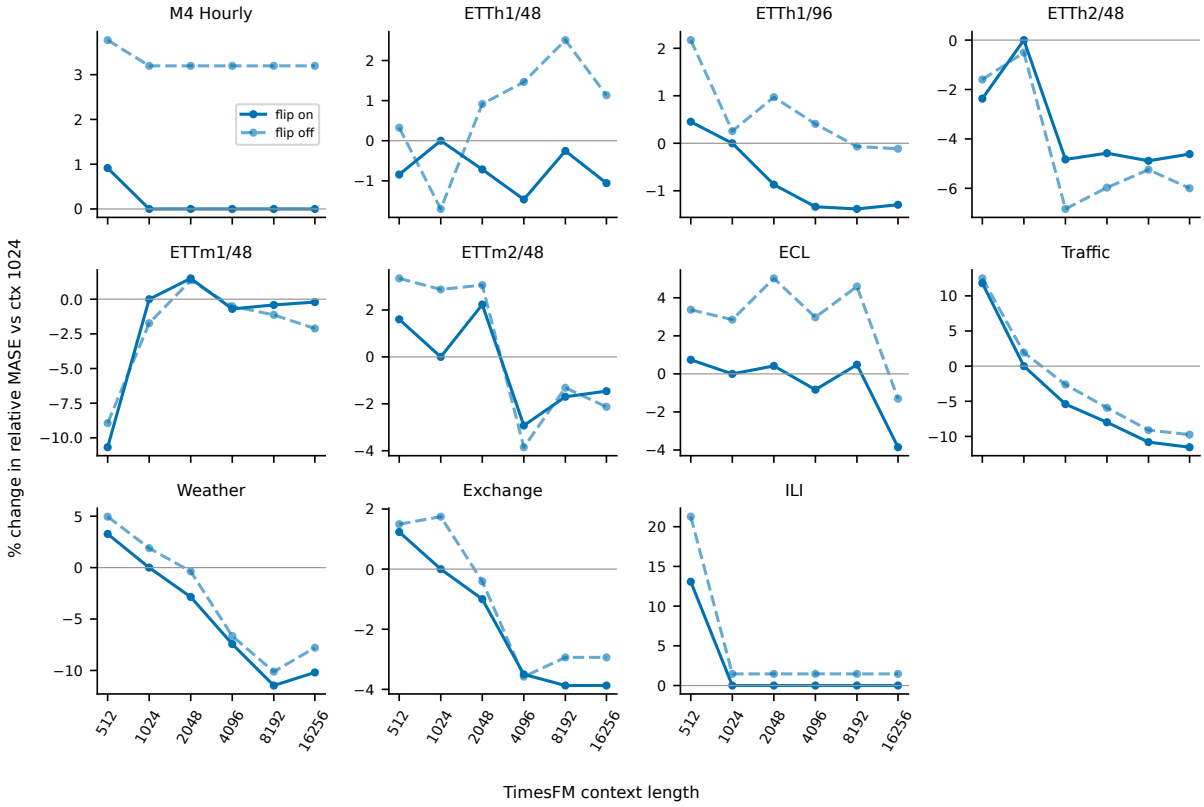


Figure 3: TimesFM context-length curves per task (% change vs ctx 1024), flip invariance on and off.

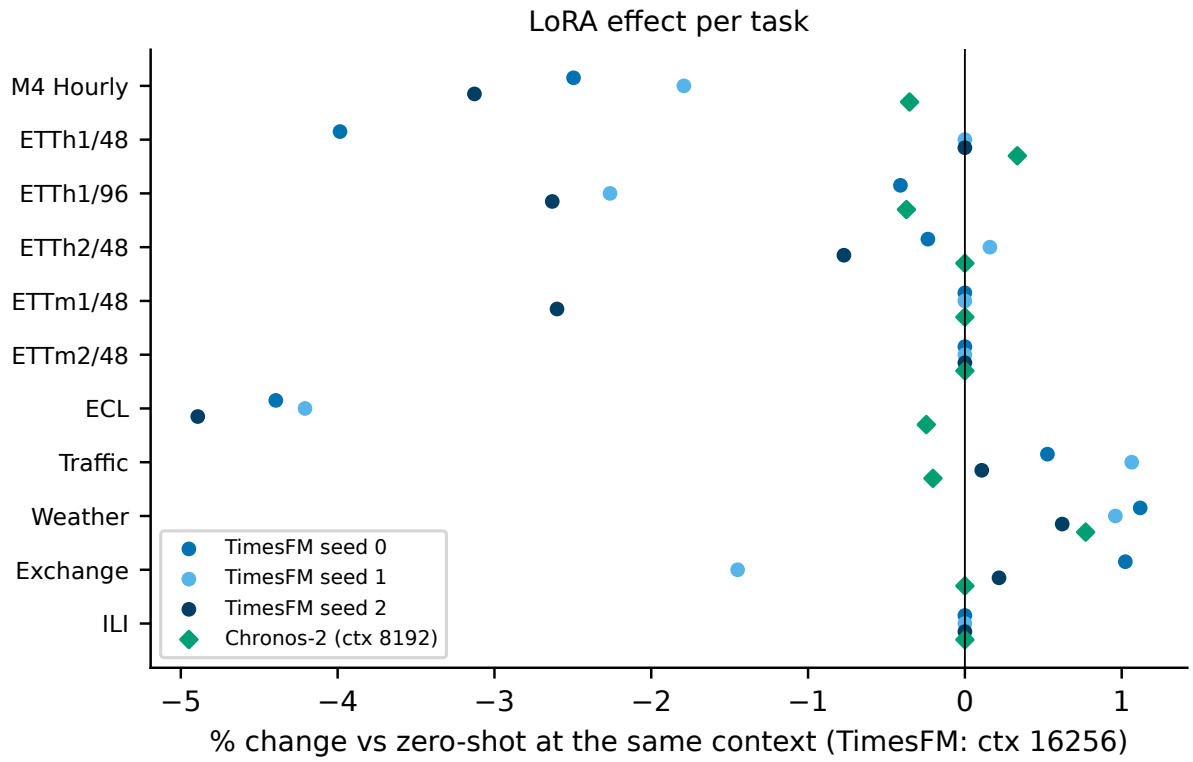


Figure 4: LoRA effect per task: three TimesFM seeds and Chronos-2.

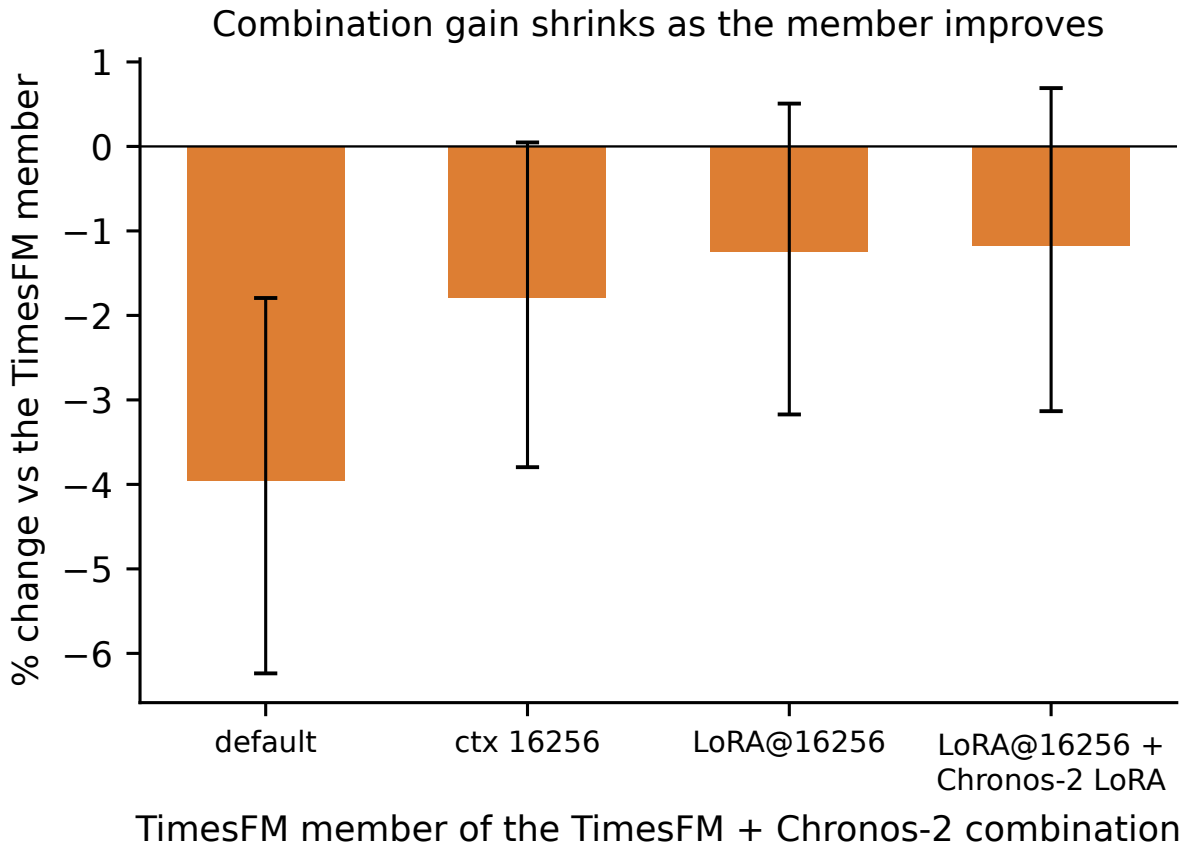


Figure 5: Combination gain with 95% intervals for successively improved TimesFM members.

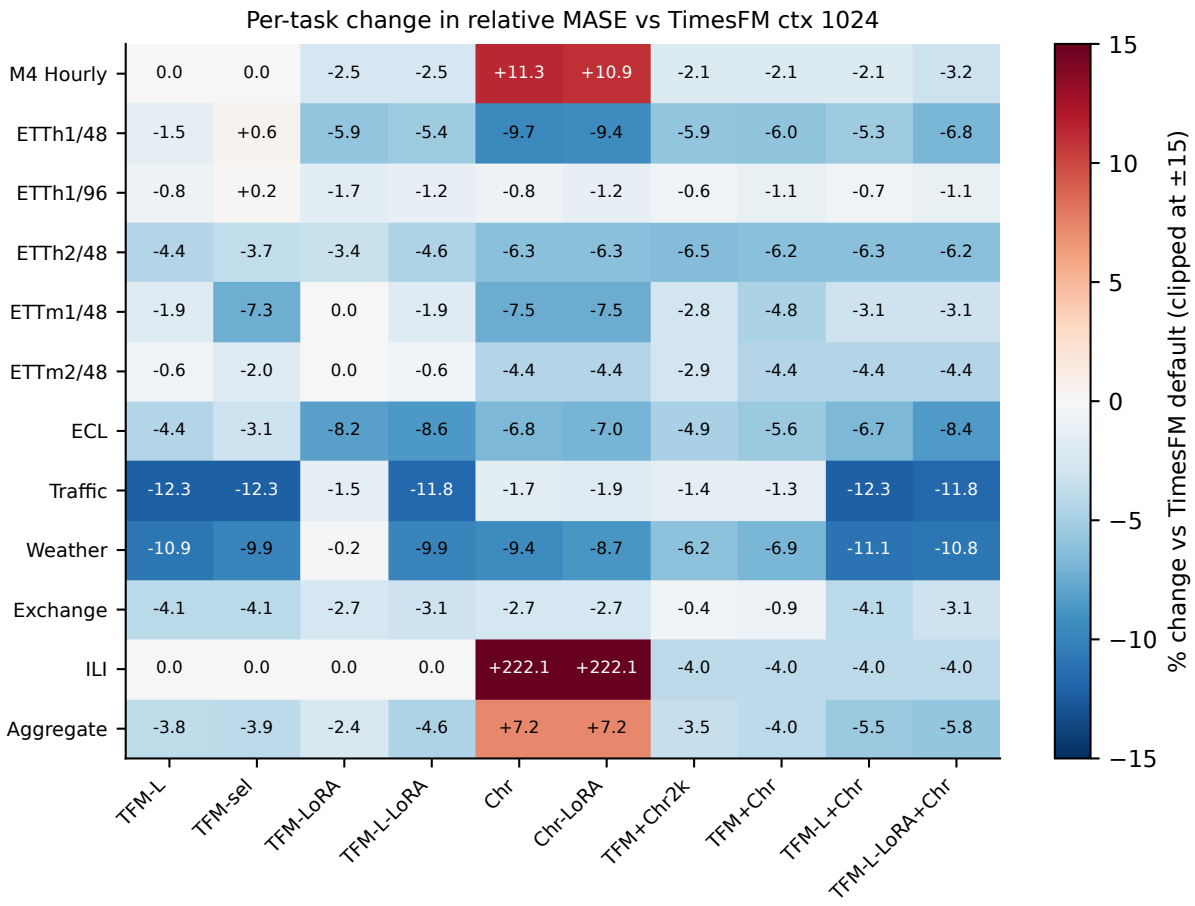


Figure 6: Per-task change vs default TimesFM for the main strategies (common evaluation set).

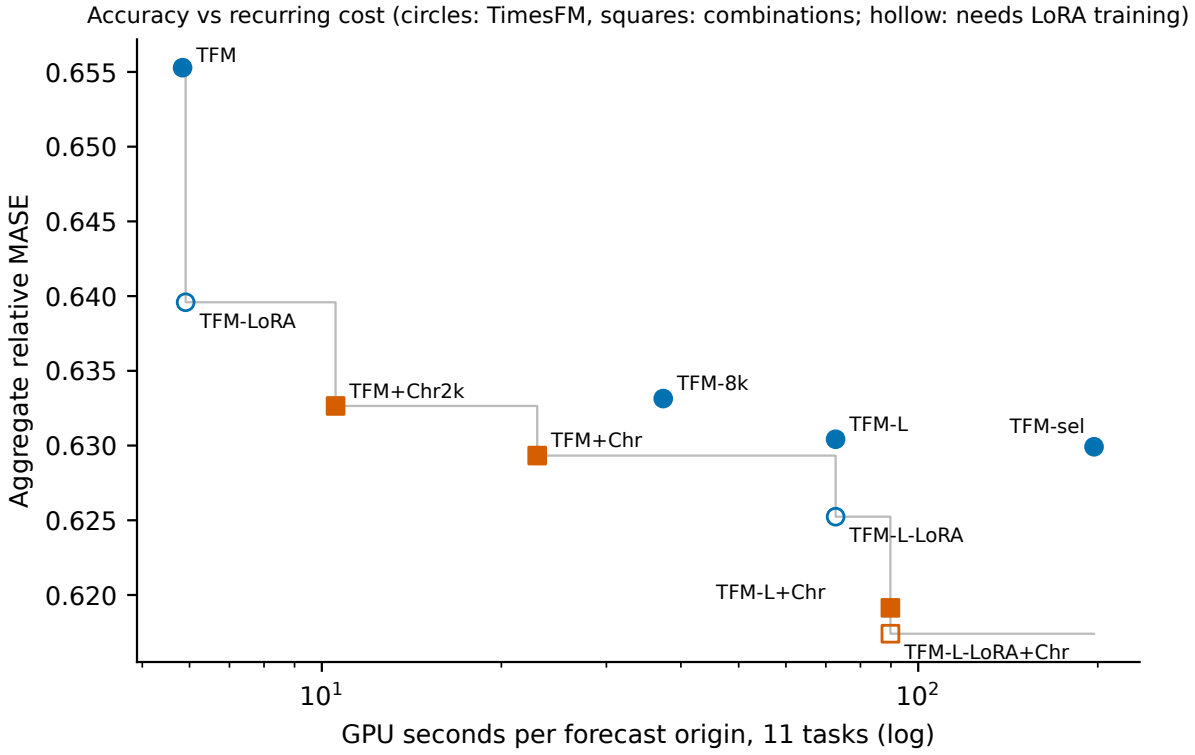


Figure 7: Accuracy vs recurring inference cost; the grey step is the cost–accuracy frontier. Chronos-2 alone (0.702, 17 s) and TabPFN-TS (0.758, ≈4,100 s) lie outside the plotted range.